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Received: 21 December 2025

Accepted: 3 February 2026

Cite this article as: Wang, Z., Duan, W., Duan, B. *et al.* 2-Ethylhexanol inhibit *Botrytis cinerea* by interfering in sugar and amino acid metabolism. *npj Sci Food* (2026). <https://doi.org/10.1038/s41538-026-00753-3>

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2-Ethylhexanol inhibit *Botrytis cinerea* by interfering in sugar and amino acid metabolism

Zipeng Wang^{1,2,5}, Wenhui Duan^{1,3,5}, Bin Duan^{2,5}, Haiwei He^{1,4}, Jiang Nan^{1,2}, Xin Yan^{1,4}, Shusheng Lei^{1,2}, Yuqi Bin¹, Chaojin Chen¹, Xiaozhen Yue¹, Xiangbin Xu³, Shuzhi Yuan¹, Jinhua Zuo¹, Jia Liu^{2,*}, Xiaodi Xu^{1,*}, & Qing Wang^{1,**}

¹Key Laboratory of Vegetable Postharvest Processing, Ministry of Agriculture and Rural Affairs, Beijing Key Laboratory of Fruits and Vegetable Storage and Processing, Institute of Agri-food Processing and Nutrition, Beijing Vegetable Research Center, Beijing Academy of Agriculture and Forestry Sciences, Beijing 100097, China. ²Chongqing Key Laboratory for Germplasm Innovation of Special Aromatic Spice Plants, College of Smart Agriculture/Institute of Special Plants, Chongqing University of Arts and Sciences, Chongqing, 402160, China. ³Key Laboratory of Food Nutrition and Functional Food of Hainan Province, School of Food Science and Engineering, Hainan University, Haikou 570228, China. ⁴College of Agriculture, Guangxi University, Nanning 530004, China. ⁵These authors contributed equally: Zipeng Wang, Wenhui Duana, Bin Duan. * Corresponding author. e-mail: jialiuliu1983@163.com; xuxiaodi@baafs.net.cn; wangqing@baafs.net.cn.

Abstract

The aim of this study was to evaluate the efficacy and underlying mechanism of 2-ethylhexanol (2-EH), a volatile compound, against the postharvest pathogen *Botrytis cinerea* on cherry tomato fruit. 2-EH demonstrated potent fungicidal activity *in vitro*, effectively inhibiting mycelial growth (100% inhibition at 0.08 mL L⁻¹) and spore germination by disrupting membrane integrity. *In vivo*, treatment significantly reduced gray mold lesions and maintained fruit quality by delaying senescence through reduced respiration and ethylene release and enhancing antioxidant enzyme activities. Transcriptomic analysis of *B. cinerea* revealed that 2-EH acts primarily by suppressing core metabolic pathways, with significant downregulation of genes related to starch/sucrose metabolism, proline and tyrosine metabolism, thereby impairing energy acquisition and stress defense. These findings establish 2-EH as a highly effective and safe volatile treatment for postharvest disease control and quality preservation of cherry tomatoes.

Keywords 2-Ethylhexanol; *Botrytis cinerea*; Cherry tomato; Postharvest control; Metabolism; Transcriptome

Introduction

The tomato (*Solanum lycopersicum* L.) is extensively grown and serves as a major dietary component across the globe¹. As a valuable dietary source, tomatoes are rich in nutrients like lycopene, carotenoids, vitamin C, dietary fiber, and other bioactive compounds². However, mature tomato fruit quickly senesce and decay during postharvest storage and transportation due to two main factors: their nature as climacteric fruits and their high susceptibility to pathogens^{3,4}. Gray mold, caused by the necrotrophic fungus *Botrytis cinerea*, is the predominant postharvest disease in tomato fruit. The fungus causes fruit decay and serious economic losses by secreting cell wall-degrading enzymes and virulence factors into fruit tissues^{5,6}. As a model necrotrophic fungus, *B. cinerea* is widespread in the environment and causes rot in various preharvest and postharvest agricultural products, aided by its capacity for quiescent infection and frequent genetic variations^{7,8}.

Various strategies have been developed to limit *B. cinerea* development and reduce postharvest losses, including chemical fungicides, physical methods, and biological control^{9,10}. For example, sulfur dioxide (SO₂), is a common chemical fungicide that inhibits *B. cinerea* on postharvest grapes. Its mechanism involves activating defense mechanisms by enhancing antioxidant capacity by promotion of reactive oxygen species (ROS) scavenging enzyme activities and the ascorbate-glutathione (AsA-GSH) cycle¹¹. Similarly, combined plasma-activated salicylic acid and in-package dielectric barrier discharge plasma (PASA-DBD) treatments have been shown to reduce gray mold incidence in strawberries and strengthen host resistance by increasing the activities of antioxidant enzymes, chitinase, and β -1,3-glucanase¹⁰. The volatile organic compound (VOC) 2-phenylethanol (2-PE), naturally emitted by plants or microorganisms, reduces gray mold incidence

in tomatoes. Its dual action involves enhancing host defense (e.g., by increasing antioxidant enzyme activities), and directly killing *B. cinerea* by triggering autophagy ROS accumulation⁶. However, the long-term application of traditional fungicides often raises environmental pollution and food safety concerns, and increases the risk of resistant strains. Moreover, the specialized equipment required for some physical methods limits their widespread application and promotion¹². Consequently, control strategies have been studied recently, with the goal of finding environment-friendly approaches that offer rapid efficacy and greater security¹³.

2-Ethylhexanol (2-EH) is a VOC naturally produced by certain microorganisms and plants. Due to its low pour point, high oxidative stability, and favorable safety profile, it is widely used in edible flavorings, biofuels, biolubricants, nonionic surfactants, and green plasticizers¹⁴⁻¹⁶. Recent studies have identified 2-EH as an effective antifungal agent. For example, it inhibits the mycelial growth of *Fusarium oxysporum* f. sp. *cucumerinum* (FOC) by decreasing ergosterol content, disrupting cell integrity, inducing ROS accumulation, and causing DNA damage¹⁷. Furthermore, 2-EH suppresses *Aspergillus flavus* growth in food and grains, significantly reducing fungal biomass and aflatoxin B1 (AFB1) production. This effect is mediated by cell membrane disruption and downregulation of ergosterol synthesis genes¹⁸. In addition, 2-EH has been shown to maintain the postharvest quality of pak choi by enhancing antioxidant enzyme activity¹⁹.

However, the effect of 2-EH on *B. cinerea* remains largely unknown. Therefore, this study aimed to investigate the inhibitory effect of 2-EH on *B. cinerea* mycelial growth and spore germination, including transcriptomic analysis of the underlying mechanism, and evaluate the efficacy of 2-EH in inducing disease resistance in tomato during postharvest storage. The results

are expected to expand the understanding of the antifungal mechanism of 2-EH and offer a novel, safe strategy to control gray mold.

Results

2-EH inhibits *B. cinerea* *in vitro* and *in vivo*

2-EH significantly inhibited the colony growth and spore germination of *B. cinerea* *in vitro*, with the inhibitory effect enhancing proportional to the increase in 2-EH concentration (Fig. 1a and c). After 72 h of incubation, the control group exhibited a colony diameter of 84.10 ± 0.03 mm. The colony diameters in 0.03 and 0.05 mL L⁻¹ 2-EH treatments were significantly reduced, corresponding to the growth inhibition rates of 26% and 38%, respectively (Fig. 1b). Notably, the highest concentration, 0.08 mL L⁻¹ 2-EH, resulted in complete inhibition (100% inhibition) of *B. cinerea* within 72 h period (Fig. 1a and b). The spore germination rate of control group was rapid, reaching 77% after 4 h of incubation and nearly 100% by 6 h (Fig. 1d). While the germination rate of 0.03 mL L⁻¹ 2-EH treatment was comparable to the control after 10 h, the germ tube length was markedly reduced to only 10% of the controls length (Fig. 1d and e). Furthermore, no spores had germinated in the 0.05 and 0.08 mL L⁻¹ 2-EH treatment groups within the 10 h-incubation period, resulting in a germ tube length of 0 μ m (Fig. 1c and e).

2-EH significantly inhibited the growth of *B. cinerea* on wounded cherry tomato fruit, with the most pronounced effect observed at the highest tested concentration, 1 mL L⁻¹ (Fig. 1f). After 72 h of treatment, the lesion diameter in the control group was 11.2 mm. The 0.5 and 1 mL L⁻¹ 2-EH treatments significantly reduced the lesion diameter to 6.05 mm and 5.33 mm, corresponding to disease inhibition rates of 46% and 52.3%, respectively (Fig. 1g).

2-EH maintains the quality of cherry tomato fruit

2-EH treatment significantly decreased the natural disease incidence of cherry tomato fruit during 16 d of storage (Fig. 2a). After this period, the decay incidence in the control group reached 21%. The 0.5 mL L⁻¹ 2-EH treatment significantly suppressed this decay to 10.14%, representing a 51.6% reduction in disease compared to the control (Fig. 2b). the treatment also delayed the decline in total soluble solids (TSS) content, maintaining a significantly higher TSS level compared to the control group after 7 d of storage (Fig. 2c). Furthermore, 0.5 mL L⁻¹ 2-EH delayed the increase in the weight loss rate (Fig. 2d). By the end of the 16 d storage period, the weight loss rate was significantly lower in the treatment group (2%) compared to the control group (4%) (Fig. 2d). Compared with the control, 2-EH treatment more effectively maintained the firmness of fruit throughout storage (Fig. 2e). Physiologically, 2-EH treatment significantly reduced both the respiration rate and ethylene release rate of cherry tomato fruit during postharvest storage (Fig. 2f and g). This reduction in metabolic activity, typical of delayed ripening, was observed throughout postharvest storage, delaying the climacteric peaks of both CO₂ and ethylene production compared to the untreated fruit.

2-EH increases the activities of antioxidant enzymes in cherry tomato fruit

The 0.5 mL L⁻¹ 2-EH treatment significantly enhanced the activities of key antioxidant enzymes in cherry tomato fruit compared to the control group throughout postharvest storage suggesting an enhanced defense response (Fig. 2h-k). The superoxide dismutase (SOD) activity followed a similar trend in both groups, increasing initially before declining, with the peak occurring at 10 d of storage. At this peak, the 2-EH treated fruit exhibited 535.31 U g⁻¹, which was 26.6% higher

than the control peak of 422.91 U g⁻¹ (Fig. 2h). The activities of catalase (CAT) and peroxidase (POD) were also significantly sustained by the treatment. By the end of 16 d of storage, CAT activity in the treatment group was 61% higher (1.27 times) and POD activity was 27% higher (1.27 times) than their respective control values (Fig. 2i and j). The ascorbate peroxidase (APX) activity gradually increased in both groups throughout the storage period. By 16 d, the treated fruit maintained a final APX activity of 916.04 U g⁻¹, approximately 7.2% higher than the control value of 854.01 U g⁻¹ (Fig. 2k). This collective increase in enzymatic defense indicates that 2-EH treatment helped the fruit better cope with oxidative stress during storage.

2-EH inhibits spore viability of *B. cinerea*

The influence of 2-EH on *B. cinerea* spore viability was assessed using the differential fluorescence indicators fluorescein diacetate (FDA) and propidium iodide (PI). FDA is hydrolyzed by intracellular esterases in viable cells, resulting in green fluorescence (viability). In contrast, PI enters spores with compromised membrane integrity (non-viable) and stains the nucleic acids red. Compared with the control, 0.08 mL L⁻¹ of 2-EH treatment drastically reduced the spore viability of *B. cinerea* (Fig. 3a). In the control group, 75% of spores were stained by FDA (viable), while only 11% were stained by FDA in 2-EH treatment group representing an 85.3% reduction in viable spores (Fig. 3b). Conversely, 85% of spores in 2-EH treatment group were stained by PI (non-viable), indicating extensive membrane damage and loss of cellular integrity, which aligns with the strong fungicidal activity observed *in vitro*. Negligible PI fluorescence was observed in the control group (Fig. 3b).

Transcriptomic analysis of the effect of 2-EH on *B. cinerea*

Exposure to 2-EH for 2 h triggered pronounced transcriptional reprogramming in the mycelium of *B. cinerea*, indicating a rapid molecular response to the volatile treatment. The sequencing data exhibited high reliability, with Q30 values exceeding 95.64% and GC contents ranging from 46.82% to 47.19% across all samples, confirming the overall quality and consistency of the dataset. Comparative transcriptome analysis between the control and the 0.08 mL L⁻¹ 2-EH treatment group identified 3,575 differentially expressed genes (DEGs), including 1,700 that were up-regulated and 1,875 that were down-regulated (Supplementary Fig. S1a and b). Principal component analysis (PCA) clearly separated the control and treated samples, highlighting distinct and reproducible transcriptional profiles between the two groups (Supplementary Fig. S1c). To further elucidate the biological significance of these transcriptional changes, the functional categories and metabolic pathways associated with the DEGs were explored through Gene Ontology (GO) annotation and Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway enrichment analyses (Supplementary Fig. S1d–k).

2-EH impacts starch and sucrose metabolism in *B. cinerea*

KEGG analysis revealed that starch and sucrose metabolism was significantly inhibited by 2-EH, with 16 DEGs identified in this pathway. A strong inhibitory effect was observed, as 15 genes, including *GYG* (*Bcin10g02280*), *GBE1* (*Bcin15g02090*), *TPS* (*Bcin07g00940*), *UGP2* (*Bcin01g03640*), *pgm* (*Bcin11g02490*), *SGA1* (*Bcin04g00030*, *Bcin04g04190*, and *Bcin12g03390*), *GBH1* (*Bcin13g02100*), *IMA* (*Bcin02g05690*), *AMY* (*Bcin03g00540*, *Bcin08g02130*, and *Bcin04g06250*), *malZ* (*Bcin11g06440* and *Bcin14g00650*), were downregulated while only *HK* (*Bcin12g00680*) was upregulated (Fig. 4a). RT-qPCR validation confirmed this downregulation

(Fig. 4b). Specifically, the expression of genes encoding *IMA* (*Bcin02g05690*) and *pgm*, which are critical for central sugar metabolism, were sharply downregulated, dropping to only 5% of the control level (Fig. 4b).

2-EH impacts proline metabolism in *B. cinerea*

Proline metabolism, critical for osmotic stress defense and as a nitrogen and carbon reserve, was another pathway strongly affected with 13 DEGs identified, including *E1.2.1.88* (*Bcin11g06270*), *PRODH* (*Bcin12g00550*), *DAO* (*Bcin13g00810*), *speB* (*Bcin13g00080* and *Bcin09g04320*), *ALDH* (*Bcin13g05810*, *Bcin02g03910*, and *Bcin03g01130*), *E4.1.1.17* (*Bcin06g02830*), *GOT1* (*Bcin01g01410*), *speD* (*Bcin02g03180*), *speE* (*Bcin14g01740*), and *FMS1* (*Bcin14g01450*). Transcriptome data indicated a complete downregulation of all the genes in this pathway in the 2-EH treated group (Fig. 5a). RT-qPCR confirmed the severe suppression of this pathway (Fig. 5c). *DAO* and *PRODH* expression levels were markedly suppressed, reaching approximately 1% and 10% of the control, respectively, thereby suggesting a near-total inhibition of proline degradation and utilization (Fig. 5c).

2-EH impacts tyrosine metabolism in *B. cinerea*

Tyrosine metabolism, essential for amino acid homeostasis and virulence factor synthesis, was also affected, with 11 DEGs identified (Fig. 5b). Among these, *ARO8* (*Bcin02g03270*) was upregulated, while 10 DEGs, including *AOC3* (*Bcin01g08900* and *Bcin02g05100*), *GOT1* (*Bcin01g01410*), *adhP* (*Bcin01g02880* and *Bcin05g07670*), *FAH* (*Bcin04g01020*), *FAHD1* (*Bcin01g05530*), *COMT* (*Bcin12g01950* and *Bcin12g03240*), and *femA* (*Bcin13g01940*), were downregulated (Fig. 5b). RT-qPCR results confirmed the overall downregulation of the pathway

(Fig. 5c), suggesting an impairment in the ability of the fungus to synthesize tyrosine-derived metabolites necessary for cellular structure and function.

Discussion

Our study confirms that 2-ethylhexanol is an effective postharvest control agent against gray mold caused by *B. cinerea*. 2-EH is naturally produced by antifungal bacteria, such as *Stenotrophomonas* sp. NAU1697 and *Lysinibacillus fusiformis*^{16,17} and has established food safety approval as a flavor additive¹⁴, highlighting its potential as a safe alternative to synthetic fungicides. We found that 2-EH rapidly inhibited both colony growth and spore germination of *B. cinerea* *in vitro*, and significantly reduced lesion expansion *in vivo* on cherry tomato fruit (Fig. 1 and 6). Spore germination is the critical first step in *B. cinerea* infection^{20,21}. The ability of 2-EH to suppress spore germination and drastically reduce spore viability by compromising the fungal membrane integrity (Fig. 3) provides the foundational explanation for its high efficacy in controlling postharvest decay.

VOCs have been shown to maintain postharvest quality of fruits and vegetables, including sweet cherry fruit, blueberry, citrus, and broccoli^{16,22-24}. Previous study reported that 2-EH fumigation can delay postharvest leaf yellowing and maintain the quality of pak choi¹⁹. We found that 2-EH fumigation effectively maintained the quality of cherry tomato fruit throughout the storage period. In addition, we further revealed 2-EH achieved this by significantly delaying senescence and enhancing intrinsic disease resistance (Fig. 2). As a climacteric fruit, tomato ripening is driven by a burst of ethylene production, which accelerates respiration and softening²⁵. 2-EH treatment successfully reduced both the ethylene release rate and the respiration rate, thereby delaying the

climacteric peak. This physiological slowdown was reflected in the maintenance of fruit quality, including higher firmness, sustained, TSS, and a reduced weight loss rate (Fig. 2).

Pathogen infection and senescence inherently increase ROS, leading to oxidative damage^{6,26}. 2-EH treatment significantly enhanced the activities of key antioxidant enzymes SOD, CAT, POD, and APX (Fig. 2). This enhanced enzymatic activity allows the fruit to more effectively scavenge excess ROS, with O_2^- catalysed by SOD and H_2O_2 destroyed by APX, CAT, and POD^{27,28}, thereby increasing the fruit's defense mechanisms against both the pathogen and oxidative stress associated with ripening.

As a crucial physiological barrier of pathogens, the cell membrane often serves as a core target for VOCs to exert their antifungal effects. VOCs can disrupt the cell membrane integrity, perturb intracellular homeostasis, and ultimately achieve the inhibitory effect on pathogens. This mode of action has been validated in studies on the antifungal activity of several VOCs against *B. cinerea*, including 3-ethyl-1-butanol, 1-octanol, and 3-Methyl pentanoic acid²⁹⁻³¹. Similarly, 2-EH can also significantly impair the cell viability of *B. cinerea* by targeting and disrupting the integrity of its cell membrane, thereby inhibiting the growth and pathogenicity of *B. cinerea*. Currently, there are limited reports on the molecular mechanisms by which VOCs inhibit *B. cinerea*. Previous studies have shown that both 2,3-butanedione and (*E*)-2-hexenal disrupt cellular homeostasis, DNA-binding, and key aspects of energy metabolism in *B. cinerea*^{32,33}. Notably, our transcriptomic analysis showed that 2-EH downregulated genes involved in starch and sucrose metabolism as well as proline and tyrosine metabolism in *B. cinerea*, suggesting energy deprivation and increased stress sensitivity. 2-EH severely targeted the starch and sucrose metabolism pathway (Fig. 4). This

pathway is crucial for carbon utilization and energy generation³⁴. We observed the downregulation 15 genes, including those critical for breaking down sugars like *malZ*, *AMY*, and *IMA*, and those involved in central glycolysis like *pgm*^{35,36}. The drastic reduction in key enzyme expression indicated by 95% reduction in *IMA* and *pgm* suggests that 2-EH effectively starves the fungus by preventing the conversion of storage and host sugars like sucrose, maltose, isomaltose into metabolizable D-glucose.

2-EH strongly downregulated genes involved in proline and tyrosine metabolism (Fig. 5). The coordinated downregulation of 13 genes involved in proline metabolism, including *PRODH* and *DAO*, implying that 2-EH likely restricts the fungus from using proline as a stress-protective molecule or as a reserve carbon/nitrogen source^{37,38}. Furthermore, the inhibition of tyrosine metabolism genes (Fig. 5b) suggests an impairment in the synthesis of important resistance substances, such as flavonoids and total phenols³⁹. The downregulation of *GOT1*, an enzyme that links amino acid and carbohydrate metabolism⁴⁰, indicates that 2-EH destroys metabolic homeostasis by isolating and preventing the cross-talk among these crucial pathways, which has rarely been documented in other microbial VOC studies.

As an effective, non-synthetic volatile, 2-ethylhexanol controls gray mold caused by *B. cinerea* and preserves the quality of postharvest cherry tomato. The fungicidal action of 2-EH is two folds (Fig. 6), first, it acts directly by disrupting the fungal cell membrane, leading to a significant reduction in spore viability and germination. Second, transcriptomic analysis revealed that 2-EH profoundly impairs core fungal metabolism by downregulating key genes in pathways critical for energy and stress management, specifically starch/sucrose and proline/tyrosine metabolism.

Furthermore, 2-EH treatment improved fruit quality by reducing respiration and ethylene release thereby delaying senescence and enhancing the fruit's innate antioxidant enzyme defense system. These results demonstrate that 2-EH is a promising and safe alternative for integrated postharvest disease control and quality maintenance.

Methods

Fruit, pathogen, and reagents

Cherry tomato fruit (*Solanum lycopersicum* cv. Fushan88) were harvested at the fully ripe (mature red) stage from a commercial farm (Beijing, China), and immediately transported to the laboratory. Fruit of uniform size and maturity, free from visible damage or disease were selected. Fruit were surface sterilized by soaking in 2% (v/v) sodium hypochlorite for 2 min. they were then rinsed thoroughly with sterile distilled water (SDW) and air dried at room temperature prior to use in the experiments.

The gray mold pathogen, *B. cinerea* strain B05.10, was maintained on potato dextrose agar (PDA) plates and cultured at 25 °C for 7-14 d. Conidia were harvested from the cultures. The suspension was filtered through two layers of sterile super filter cloth to remove mycelial fragments. The spore concentration was determined using an automated cell counter (Countstar Mira FL Pro), and subsequently adjusted.

2-Ethylhexanol (CAS. NO. 104-76-7) with a purity of 98% was purchased from Macklin (Beijing, China). Sodium hypochlorite solution (CAS. NO. 7681-52-9) with an active chlorine basis of 6-14% was purchased from Macklin (Beijing, China). Fluorescein diacetate (FDA) staining solution (CAS No. 596-09-8) with a concentration of 10 mg mL⁻¹ was purchased from

Coolaber (Beijing, China). Propidium iodide (PI) staining solution (CAS. NO. 25535-16-4) with a concentration of 1 mg mL^{-1} was purchased from Coolaber (Beijing, China).

***In vitro* antifungal activity of 2-Ethylhexanol**

The antifungal effect of 2-ethylhexanol on *B. cinerea* mycelial growth and spore germination were evaluated by volatile fumigation following the method⁴¹, with modifications. Mycelial plugs taken from the edge of a *B. cinerea* culture were placed in the center of fresh PDA plates. Different volumes of 2-EH were pipetted onto a single layer of sterile filter paper ($15 \times 15 \text{ mm}$) affixed to the inside of the Petri dish lid, resulting in final concentrations of 0.03, 0.05, and 0.08 mL L^{-1} of 2-EH within the plate. The untreated control received an equal volume of SDW. The plates were immediately sealed with Parafilm and incubated at 25°C in the dark. Colony diameter was measured at 24, 48, and 72 h of incubation. The experiment was conducted with three independent biological replicates, with six Petri plates as technical replicates per treatment concentration.

Spore suspension ($1 \times 10^6 \text{ spores mL}^{-1}$) was uniformly spread onto PDA plates overlaid with a sterile cellophane membrane to facilitate subsequent microscopic observation. The plates were sealed and fumigated using the same 2-EH concentrations as described for the mycelial growth assay. After incubation at 25°C in the dark for 4, 6, 8, and 10 h, the cellophane membrane was removed and placed on a glass slide for observation under a light microscope. A spore was considered germinated when the germ tube length exceeded the spore diameter. Spore germination percentage was calculated by counting at least 200 spores from random fields of view per plate. Germ tube length was measured using image analysis software (ImageJ NIH, USA). The experiment was performed using three independent biological replicates.

***In vivo* control efficacy of 2-ethylhexanol on cherry tomato gray mold**

Cherry tomato fruit was surface wounded at the equatorial region using a 2 mm diameter sterile punch to create a standardized wound (2 mm diameter $\times$ 2 mm depth). Each wound was inoculated with the mycelial plug of *B. cinerea*. The inoculated fruit were subject to volatile treatment inside sterilized 9 L plastic boxes sealed with 0.03 mm polyethylene bags containing different concentrations of 2-ethylhexanol. The treatment concentrations were 0 mL L⁻¹ (SDW control), 0.5 mL L⁻¹ and 1 mL L⁻¹. Twenty inoculated fruit were placed in each box, ensuring no contact between fruit. Humidity (80-90%) was maintained using moist filter paper placed at the base of the box. The boxes were stored at 25 °C for the duration of the experiment. Lesion diameters on fruit were recorded at 24, 48, and 72 h after inoculation. The experiment was conducted with three independent biological replicates, with 20 fruit per replicate.

Effect of 2-Ethylhexanol on quality of cherry tomato fruit

Natural disease incidence

According to the previous method⁴¹, surface sterilized cherry tomato fruit (70 fruit per replicate/group) was stored in sealed and sterilized 9 L plastic boxes. Fruit were exposed to 2-EH (and the sterile water control). Humidity was maintained at 80–90% using moist filter paper. The fruits were stored at 25 °C for 16 d. Sampling for quality analysis and physiological assays occurred at 0, 7, 10, 13, and 16 d. Fruit tissue was rapidly frozen in liquid nitrogen and stored at -80 °C. Natural disease incidence was calculated as the percentage of fruit exhibiting disease symptoms among the total fruit. The experiment used three independent biological replicates, with 70 fruit per replicate.

Quality evaluation of cherry tomato fruit

Color (L^* , a^* , and b^*) was at three equidistant points along the cherry tomato equator using a color difference meter (CR-400, Konica Minolta). Chroma (C^*) was calculated according to values of a^* and b^* , following equations to determine C^* ⁴².

$$\text{Chroma } (C^*) = \sqrt{a^{*2} + b^{*2}} \quad (1)$$

TSS content was measured using a hand-held refractometer (WY032R; Chengdu Optical Instrument Factory, China). Results were expressed as Brix using 1 mL of filtered tomato juice. Weight loss was calculated based on the difference between the initial weight and the weight at each sampling point⁴³. The fruit firmness was measured at two opposite points along the fruit equator using a GY-40J digital fruit hardness tester (Zhejiang Top Instrument Co., Ltd., Hangzhou, China)³⁶. Results were expressed as Newtons (N).

Respiration rate and ethylene release rate

According to the previously reported methods¹⁹, respiration rate was determined by placing 30 fruits in a volume closed container for 1 h at the storage temperature (25 °C). The CO₂ concentration increase in the headspace was measured using a portable respiration analyzer (GXH-3051, Junfang Institute of Scientific Instruments, Beijing, China). Results were expressed as mg CO₂ kg⁻¹ h⁻¹.

Ethylene release rate was measured by incubating 30 fruits in an airtight container for 1 h⁴⁴. A 1 mL headspace gas was extracted and analyzed by a gas chromatograph (7820 A, Agilent Technologies, Inc., Santa Clara, USA). Ethylene release rate was expressed as mL kg⁻¹ h⁻¹. Each experiment had biological triplicates.

Activities of SOD, CAT, POD, and APX

Frozen powdered fruit samples (0.2 g) were homogenized in 1 mL of cold extraction solution, the composition of which was tailored for the assay of SOD, CAT, and POD or APX. Following centrifugation of the homogenates at 4 °C ($8000 \times g$ for 10 min for SOD, CAT, and POD; $13,000 \times g$ for 20 min for APX), the supernatants were collected as crude enzyme extracts and kept on ice for subsequent analysis. The activities of SOD, CAT, POD, and APX were measured using commercial kits from Suzhou Kemin Biotechnology Co., Ltd (China). SOD activity was measured by monitoring the absorbance at 450 nm and expressed as U g⁻¹ FW. CAT activity was measured by monitoring the absorbance at 240 nm and expressed as nmol min⁻¹ g⁻¹ FW. POD activity was measured by monitoring the absorbance at 470 nm and expressed as U g⁻¹ FW. APX activity was measured by monitoring the absorbance at 290 nm and expressed as nmol min⁻¹ g⁻¹ FW. Each experiment had biological triplicates.

Assessment of spore viability

Spore viability of *B. cinerea* following 2-EH exposure was measured using fluorescence microscopy and differential staining⁴⁵. A 5 mL PDB culture containing 5×10^6 spores mL⁻¹ of *B. cinerea* were treated with 2-EH 0.08 mL L⁻¹ or SDW (control). Cultures were incubated for 6 h at 25 °C with shaking. Spores were collected by centrifugation ($5000 \times g$, 5 min). The pellet was resuspended in phosphate-buffered saline and stained with FDA at 50 µg mL⁻¹ and PI at 20 µg mL⁻¹ for 30 min in the dark. The stained spores were washed twice in PBS via centrifugation to remove excess dye. Spores observed under a fluorescence microscope (LSM 880, Zeiss, Oberkochen, Germany) with appropriate filters (FDA staining viable spores green and PI staining non- viable

spores red). At least 200 spores were counted per replicate, and viability was calculated as the percentage of FDA-stained spores (viable). The experiment was performed using three independent biological replicates.

Transcriptome sequencing and analysis

Mycelial plugs were incubated on PDA plates covered with cellophane at 25 °C for 7 d. The cultures were then exposed to 2-EH volatile fumigation at 0.08 mL L⁻¹ (treatment) or SDW (control). Mycelium was harvested from the cellophane membrane after 2 h of treatment, flash-frozen in liquid nitrogen, and then stored at -80 °C until RNA extraction. Total RNA was extracted from three biological replicates per treatment group using the Tiangen RNA Isolation Kit (Tiangen Biotech Co., Ltd., China). cDNA library preparation and sequencing were performed by Biomarker Technology Co., Ltd. (Beijing, China) using the Illumina NovaSeq 6000 platform. Reads were mapped to the *B. cinerea* reference genome (ASM14353v4). Gene expression levels were quantified using the Fragments Per Kilobase of transcript per Million mapped (FPKM) method. Differential expression analysis was performed using the DESeq2 package. DEGs were defined by the thresholds Fold Change ≥ 2 (up- or down-regulated) and adjusted *P* value (FDR) < 0.01 . Functional annotation and enrichment analysis of DEGs were performed using GO analysis and KEGG pathway analysis.

Analysis of reverse transcription and quantitative PCR (RT-qPCR)

Total RNA was isolated from the frozen mycelium using the Vazyme RNA Isolation Kit (Vazyme Biotech Co., Ltd. Nanjing, China) following the manufacturer's instructions. Reverse transcription (RT) was performed on 500 ng of total RNA using the RT Supermix Kit (with gDNase) (Vazyme

Biotech Co., Ltd. Nanjing, China) to synthesize first-strand cDNA. Quantitative PCR (qPCR) was performed on a Real-Time PCR System (Model CFX96, Bio Rad, USA) using the SuperReal Premix Plus (SYBR Green) Kit (Vazyme Biotech Co., Ltd. Nanjing, China) in a 15 μ L reaction volume.

The *B. cinerea* actin gene (*BcactA*) gene served as the endogenous reference gene for normalization. Relative gene expression levels were calculated according to the description of the previous method³⁶. Three independent biological replicates were used for the experiment with three technical replicates. The sequences of the gene-specific primers are listed in Supplement table 1.

Statistical analysis

All data were subjected to analysis of variance (ANOVA) using SPSS Statistics 25.0 software (SPSS, Chicago, IL, USA). Data were analyzed by one-way ANOVA, followed by Tukey's multiple comparison test for post-hoc analysis. Significance was established at $P < 0.05$. For the two groups data, a Student's *t*-test was used to assess significant differences ($P < 0.05$) between the means of two treatment groups. Data visualization and the calculation of standard deviation and error bars were performed using GraphPad Prism 9.0 software (GraphPad, San Diego, CA, USA). Different lowercase letters indicate significant differences between the groups.

Data availability

The datasets used during the current study are available from the corresponding author on reasonable request.

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Funding declaration

This work was supported by Special innovation ability construction fund of Beijing Academy of Agricultural and Forestry Sciences (KJCX20251006), Special innovation ability construction fund of Beijing Academy of Agricultural and Forestry Sciences (KJCX20240508), China Agriculture Research System of MOF and MARA (CARS-23), the National Natural Science Foundation of China (32472798, 32302614), Collaborative Innovation Program of the Beijing Vegetable Research Center (XTCX202302, XTCX202301), Special innovation ability construction fund of Beijing Academy of Agricultural and Forestry Sciences (KJCX20240334, KJCX20251102).

Author contributions

Writing – review & editing, Writing – original draft, Data curation, Conceptualization: W.Z. Original draft, Methodology, Investigation, Data curation: D.W. Methodology, Investigation, Data curation: D.B. Software, Data curation: H.W. Software, Methodology: N.J. Visualization, Data curation: Y.X. Resources, Formal analysis, Conceptualization: L.S. Formal analysis, Data curation: B.Y. Formal analysis, Data curation: C.C. Investigation, Formal analysis: Y.X. Resources, Methodology: X.X. Formal analysis, Conceptualization: Y.S. Resources, Data curation: Z.J. Writing – review & editing Resources, Formal analysis: L.J. Supervision, Resources, Methodology, Formal analysis: X.X. Writing – review & editing, Validation, Supervision, Project administration, Funding acquisition, Conceptualization: W.Q.

Competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Figure Legends

Fig. 1 Effect of 2-EH on colony morphology (a), colony diameter (b), microscopic view of spores (c), percentage of spore germination (d), germ tube length (e) of *B. cinerea*. The symptoms of gray mold on cherry tomato fruit (f), lesion diameter (g). Scale bar = 50 μm . Vertical bars represent standard deviations of the means, $n = 3$. The different letters indicate significant differences among the plants according to Tukey's multiple comparison test ($P < 0.05$). The figures were created using the professional software GraphPad Prism.

Fig. 2 Effect of 2-EH on the postharvest quality of cherry tomato fruit. The phenotypic characterizations of cherry tomato fruit during storage at room temperature (a), The natural disease incidence (b), TSS content (c), weight loss (d), firmness (e), ethylene release rate (f) respiration rate (g), SOD activity (h), APX activity (i), CAT activity (j), POD activity (k) of tomato fruit. Vertical bars represent standard deviations of the means, $n = 3$. Asterisks indicate significant differences between treatment groups at $P < 0.05$ (*), $P < 0.01$ (**), $P < 0.001$ (***). The figures were created using the professional software GraphPad Prism.

Fig. 3 Effect of 2-EH on the spore viability of *B. cinerea*. Staining of *B. cinerea* spore with fluorescein diacetate and propidium iodide (a). Scale bar = 50 μm . The viable rate of *B. cinerea* spores stained by fluorescein diacetate and propidium iodide (b). Vertical bars represent standard deviations of the means, $n = 3$. Asterisks indicate significant differences between treatment groups at $P < 0.001$ (***). The figures were created using the professional software GraphPad Prism.

Fig. 4 Effect of 2-EH on starch and sucrose metabolism pathways in *B. cinerea*. Differentially

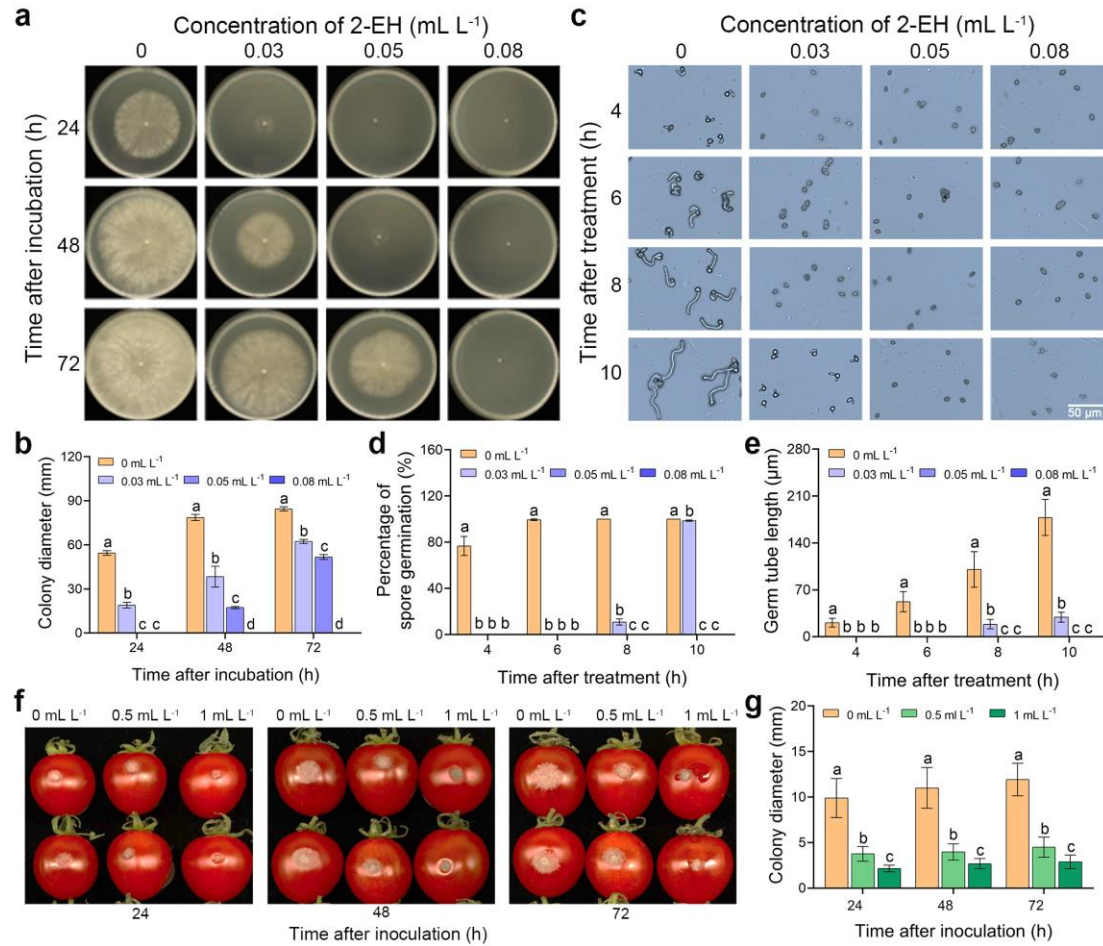
expressed genes (DEGs) enriched in KEGG pathways associated with starch and sucrose metabolism in the 0.08 mL L⁻¹ of 2-EH treatment group vs. CK comparison (a). RT-qPCR analysis of genes related to starch and sucrose metabolism pathways (b). Vertical bars represent standard deviations of the means, n = 3. Asterisks indicate significant differences between treatment groups at $P < 0.05$ (*), $P < 0.01$ (**), $P < 0.001$ (***). *GYG*, glycogenin; *GBE1*, 1,4-alpha-glucan branching enzyme; *TPS*, trehalose 6-phosphate synthase complex regulatory subunit; *UGP*, UTP-glucose-1-phosphate uridylyltransferase; *pgm*, phosphoglucomutase; *SGA1*, glucoamylase; *GBH1*, cellulose 1,4-beta-cellobiosidase; *IMA*, oligo-1,6-glucosidase; *AMY*, Alpha-amylase; *malZ*, alpha-glucosidase; *HK*, hexokinase. The figures were created using the professional software GraphPad Prism and Adobe Photoshop.

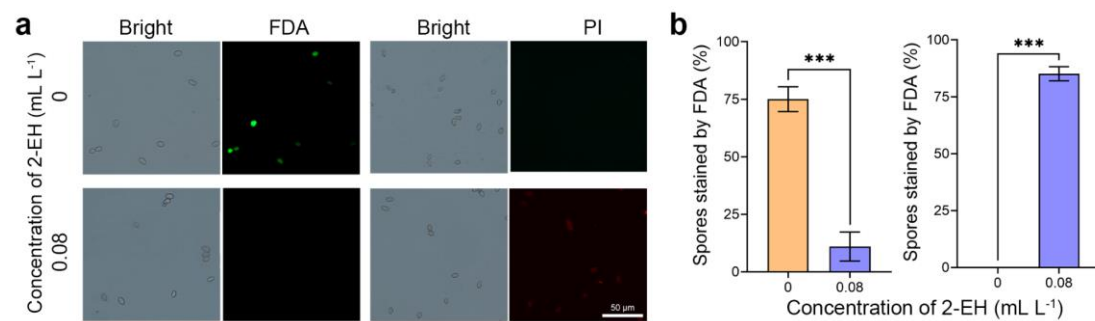
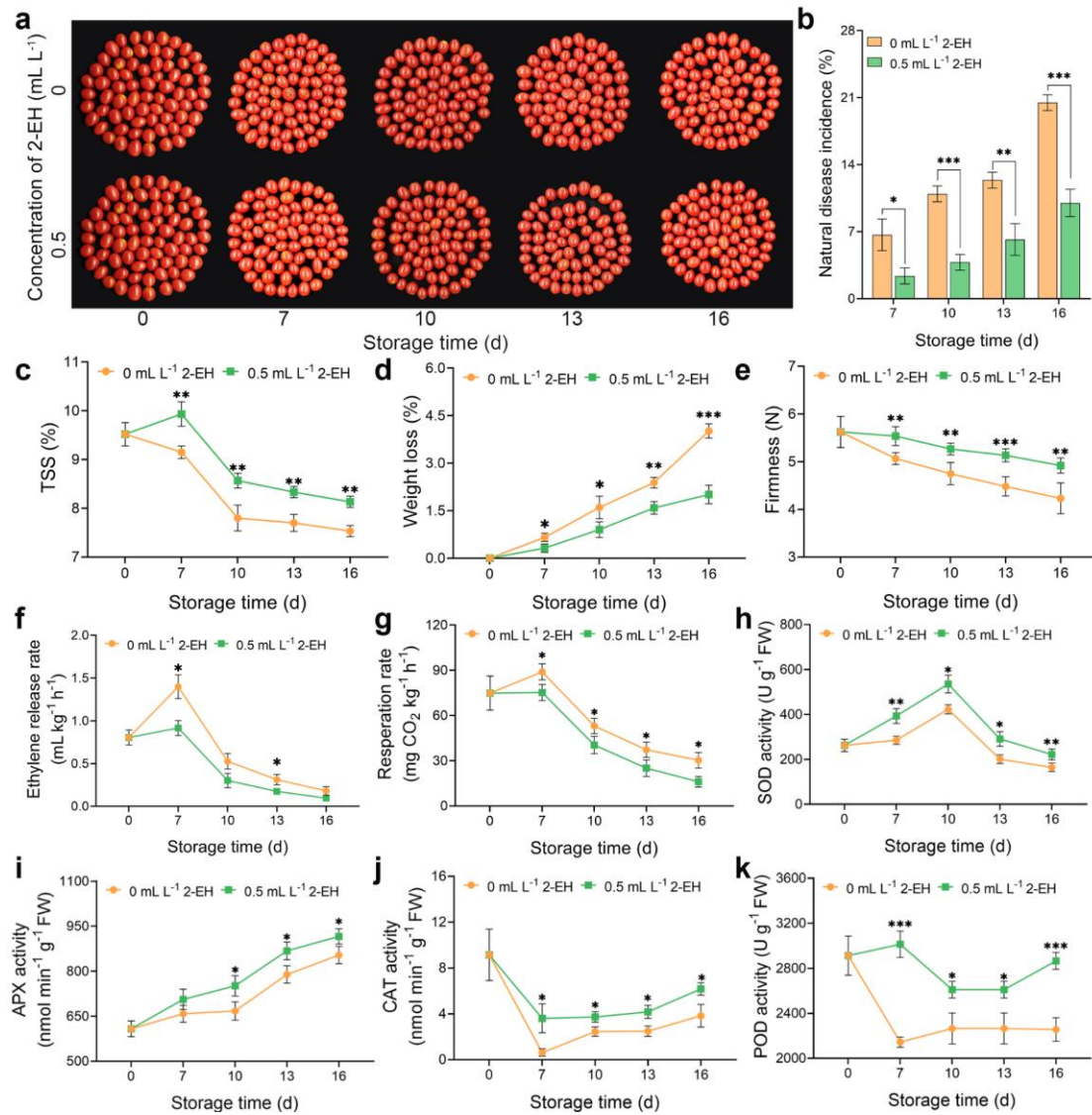
Fig. 5 Effect of 2-EH on proline and tyrosine metabolism pathways in *B. cinerea*. Differentially expressed genes (DEGs) enriched in KEGG pathways associated with proline metabolism (a) and tyrosine metabolism (b) in the 0.08 mL L⁻¹ of 2-EH treatment group vs. CK comparison. RT-qPCR analysis of genes related to proline and tyrosine metabolism pathways (c). Vertical bars represent standard deviations of the means, n = 3. Asterisks indicate significant differences between treatment groups at $P < 0.05$ (*), $P < 0.01$ (**), $P < 0.001$ (***). *E1.2.1.88*, 1-pyrroline-5-carboxylate dehydrogenase; *PRODH*, proline dehydrogenase; *DAO*, D-amino-acid oxidase; *speB*, agmatinase; *ALDH*, aldehyde dehydrogenase; *E4.1.1.17*, ornithine decarboxylase; *GOT1*, aspartate aminotransferase; *speD*, S-adenosylmethionine decarboxylase; *speE*, spermidine synthase; *FMS1*, polyamine oxidase; *ARO8*, aromatic amino acid aminotransferase; *I/2-aminoadipate transaminase*; *AOC3*, primary-amine oxidase; *GOT1*, aspartate aminotransferase;

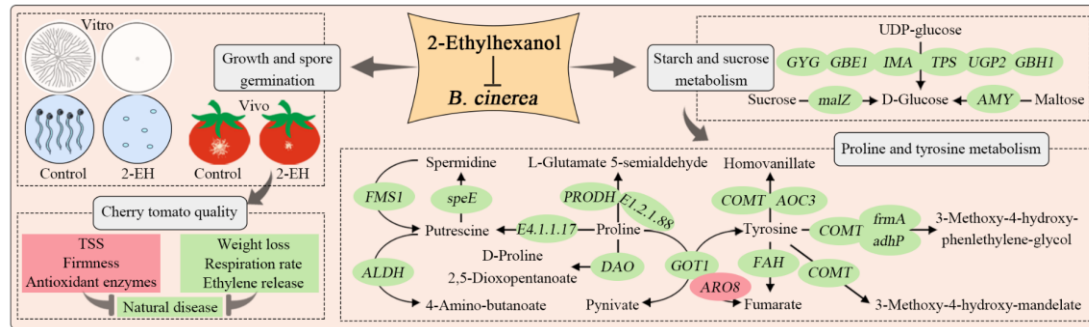
adhP, alcohol dehydrogenase; *FAH*, fumarylacetoacetase; *FAHD1*, acylpyruvate hydrolase; *COMT*, catechol O-methyltransferase; *femA*, S-(hydroxymethyl)glutathione dehydrogenase/alcohol dehydrogenase. The figures were created using the professional software GraphPad Prism and Adobe Photoshop.

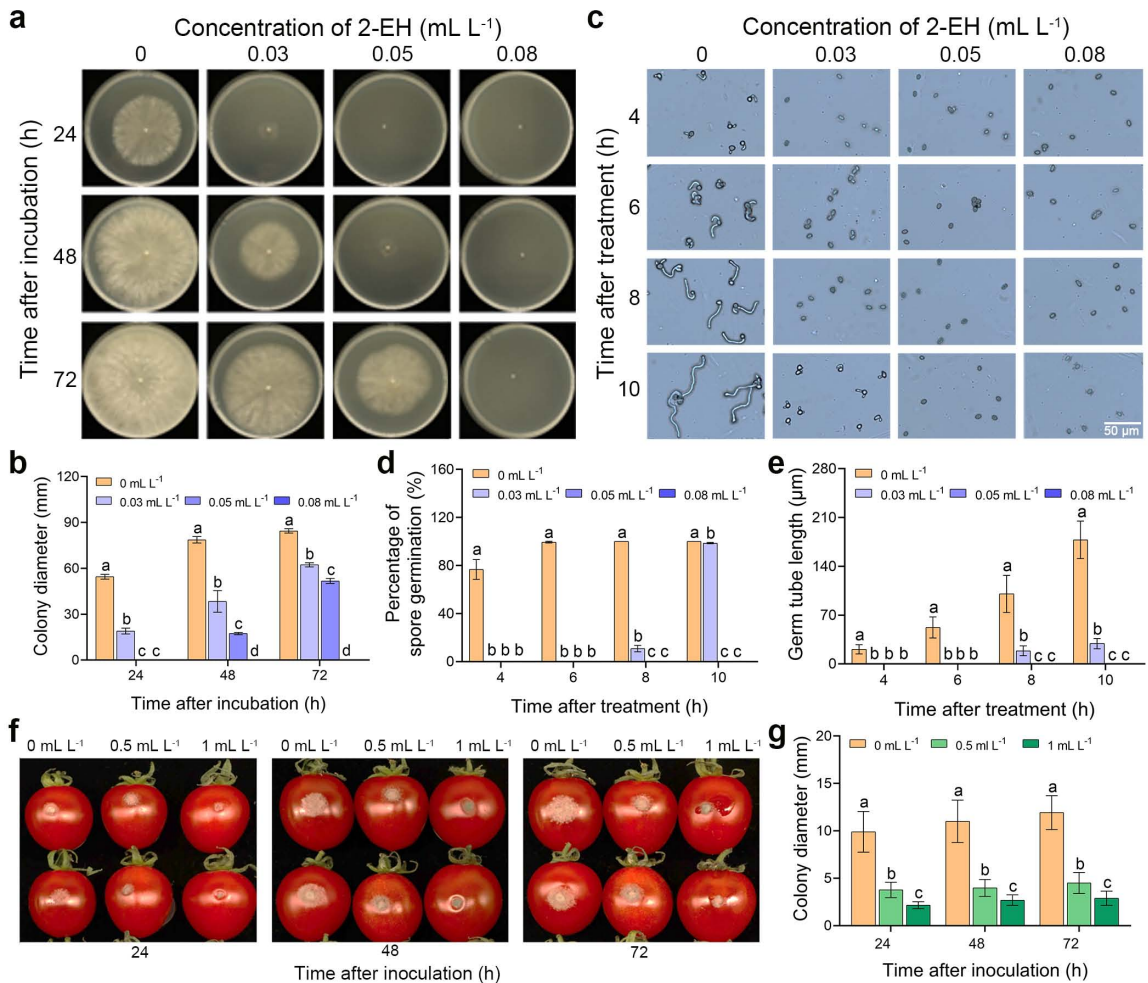
Fig. 6 A simplified model of the regulatory mechanism of the inhibition of *B. cinerea* by 2-EH.

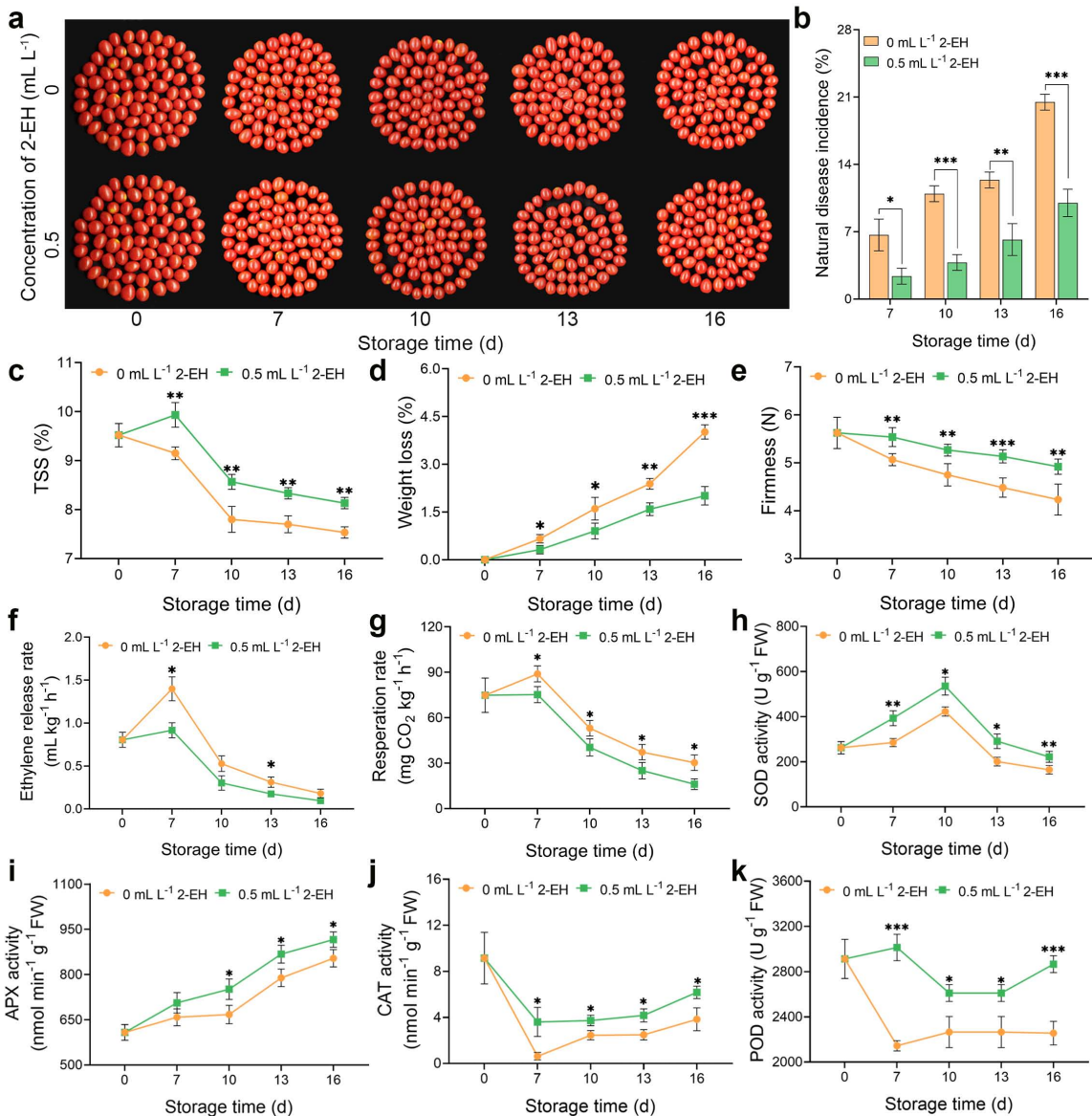
The figures were created using the professional software Adobe Photoshop.





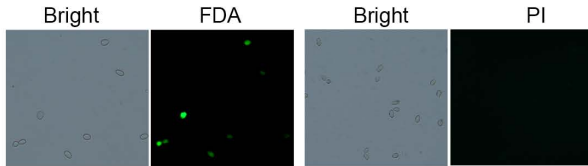




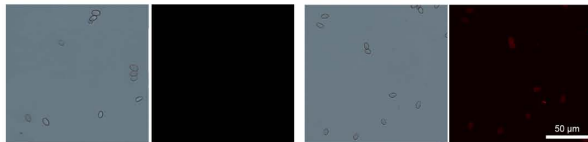
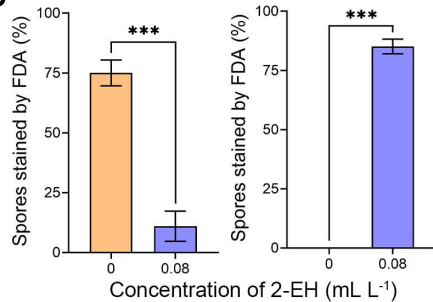


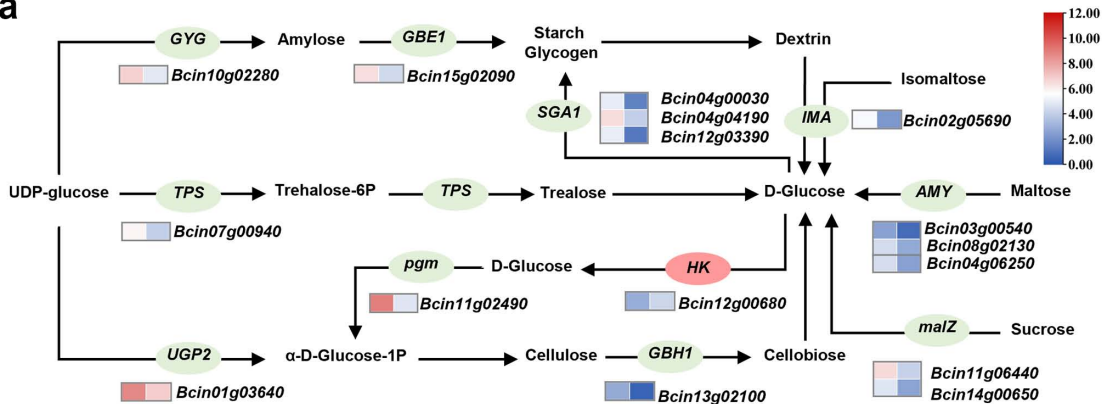
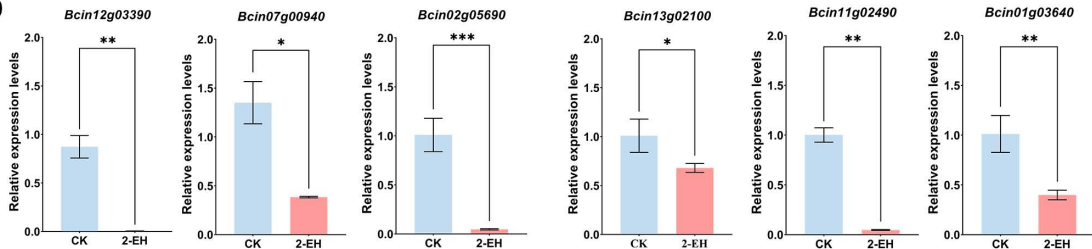
aConcentration of 2-EH (mL L^{-1})

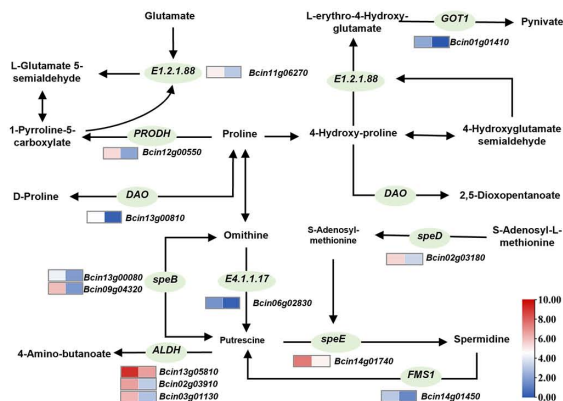
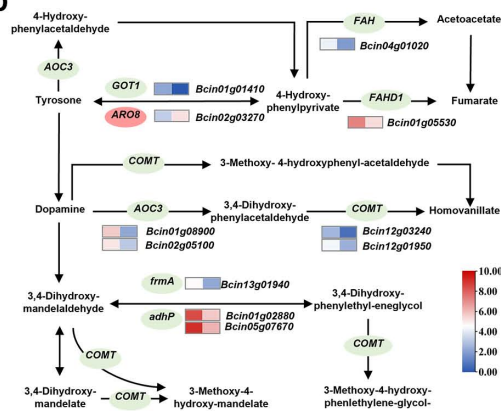
0



0.08

**b**

a**b**

a**b****c**